STA301 – Statistics and Probability

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1. Explain Null and alternative hypothesis.
2. Relation between critical region and testing?
3. Athematic mean uses?
4. Mean deviation for ungroup data was given?
5. Coefficient of variations?
6. First two moments?
7. P(A/B) and P(B/A)? Values was given
8. In least square regression line, X is a NON-RANDOM variable, whereas Y is a RANDOM variable. For example, consider the case of agricultural experiments.
9. A hypothetical population can be defined as the aggregate of all the conceivable ways in which a specified event can happen.
10. Quantitative variable
11. Which is not dispersion?
12. Census is primary data or secondary data?
13. John Tukey (1977)
14. Describe the Stem-and-Leaf Display.
15. Right-skewed distributions?
16. The positively skewed curve
17. Conditional probability of the event B given that event A has occurred, written as P(B/A), is defined by
18. Median of -3, 3, 5,6,7,10,11 is
19. If a data of 20 values, all are 5 then arithmetic mean is?
20. Which is not measure of central tendency
21. Commutative frequency distribution is also known as?
22. Law equations
23. Linear regression line
24. Y=a=bx in this equation b constat represent?
25. How to find the median?
26. Cartesian product definition and 2 sets where are also given there
27. Draw a polygon of the given data
28. Find the first 2 moments from the mean of the given data?
29. What is BCR?
30. What is hypergeometric?
31. How to find Union & Intersection?
32. When P and N given, fine mean and standard deviation.
33. Binomial distribution?
34. How many black card, diamond card, heart card, face card, queen card?
35. Continuous random topic (integration).
36. What is T-distribution?
37. Interval scale and ratio scale with examples.
38. Describe Playing cards.
39. Define Stem leaf.
40. Binomial experiment properties?
41. Anova table?
42. Binomial distribution?
43. How to find mean?
44. Find Cov(x,y) if values given.
45. How to independent prove X and Y if values given?
46. P1, N1, P2, N2 values given, find population proportion?
47. Null and alternative hypothesis given, find C.R?
48. Confidence interval, S square and N values given, find interval of sigma square?
49. What about to Z distribution?
50. Poisson distribution?
51. Mean of Hypergeometric distribution
52. Normal distribution (high priority)
53. Binomial distribution
54. Chi distribution (high priority)
55. Sampling
56. Testing hypothesis

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57. Confidence interval
58. Degree of freedom
59. F-distribution, t-test, z-test, f-test
60. Formula of z test
61. Formula n sample without replacement
62. Definition of Statistics?
63. Definition of Probability?
64. Hypothesis by z test
65. Estimation
66. Fertile deviation
67. Critical region
68. Sample distribution of sample mean
69. A table given, in which x or p(x) values given, find mean and variance? E(x) and var
70. Exhaustive event
71. Sampling errors?
72. Parameters of F distribution?
73. How to find Arithmetic mean?
74. How to find Mean or variance
75. Properties of hypergeometric?
76. Construct polygon
77. Find e3x+1
78. Find confident limit
79. Find coefficient of variance?
80. Suppose X is a variable between integers 1 to 14, what is the probability of X to b even number? (question like this)
81. About combination?
82. What is the probability of committing to be distributed among 10 persons?
83. What are the different measures of dispersion that don't use all values?
84. What are the two suits that can be black in playing cards?

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Answers:

1. **Explain Null and Alternative Hypothesis**
   - **Null Hypothesis (H_o):** A statement assuming no effect or difference exists. For example, H0:μ=50H_o: \mu = 50 means the population mean is 50.
   - **Alternative Hypothesis (H_1):** Opposes H0H_o, suggesting an effect or difference exists. For example, H1:μ≠50H_1: \mu \neq 50.
2. **Relation Between Critical Region and Testing**
   - The **critical region** is the range of test statistic values where the null hypothesis is rejected. It depends on the **significance level (α)**, such as 0.05. If the calculated statistic lies in the critical region, we reject H0H_o.
3. **Arithmetic Mean Uses**
   - The arithmetic mean represents the central value of data. It is used to compare datasets and is widely applied in economics, research, and education. It is sensitive to outliers.
4. **Mean Deviation for Ungrouped Data**
   - Formula: Mean Deviation=∑|x−x¯|n\text{Mean Deviation} = \frac{\sum |x - \overline{x}|}{n} xx = data points, x¯\overline{x} = mean, nn = number of observations. It measures the average distance between each data value and the mean.
5. **Coefficient of Variation (C.V.)**
   - Formula: C.V.=Standard DeviationMean×100C.V. = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100 It compares the variability of datasets with different units or means. Higher C.V.C.V. indicates greater relative dispersion.
6. **First Two Moments**
   - **First Moment:** Mean of the dataset, representing the central location.
   - **Second Moment:** Variance, measuring data spread. Formula: Variance=∑(x−x¯)2n\text{Variance} = \frac{\sum (x - \overline{x})^2}{n}
7. **P(A|B) and P(B|A)**
   - **Conditional Probability Formula:** P(A|B)=P(A∩B)P(B), P(B|A)=P(A∩B)P(A)P(A|B) = \frac{P(A \cap B)}{P(B)}, \, P(B|A) = \frac{P(A \cap B)}{P(A)} These values depend on the specific probabilities provided.
8. **Least Square Regression Line**
   - In regression, XX is non-random (independent variable) while YY is random (dependent variable). For example, in agricultural experiments, crop yield (YY) depends on fertilizer amount (XX).
9. **Hypothetical Population**
   - A population of all conceivable outcomes of an event, e.g., all possible outcomes of rolling a die infinitely.
10. **Quantitative Variable**
- Variables measured numerically, like age, weight, or income.
11. **Which is Not Dispersion?**
- Central tendency measures like **mean** and **median** are not measures of dispersion.
12. **Census: Primary or Secondary Data?**
- **Census data** is **primary data**, as it involves direct collection.
13. **John Tukey (1977)**
- Introduced exploratory data analysis techniques, like the **box plot**.
14. **Stem-and-Leaf Display**
- A method to organize numerical data. The "stem" represents the leading digit(s), and the "leaf" represents the trailing digit(s).
15. **Right-Skewed Distributions**
- Most data points lie on the left, and the tail extends to the right. The mean is greater than the median.
16. **Positively Skewed Curve**
- Similar to a right-skewed distribution, indicating data concentration towards the lower end.
17. **Conditional Probability P(B|A)**
- Defined as: P(B|A)=P(A∩B)P(A)P(B|A) = \frac{P(A \cap B)}{P(A)}
18. **Median of -3, 3, 5, 6, 7, 10, 11**
- Arrange: -3, 3, 5, 6, 7, 10, 11. Median = middle value = **6**.
19. **If All 20 Values Are 5, Arithmetic Mean?**
- Mean = ∑xn=20×520=5\frac{\sum x}{n} = \frac{20 \times 5}{20} = 5.
20. **Which Is Not a Measure of Central Tendency?**
- Measures of central tendency include mean, median, and mode. **Standard deviation** is not.
21. **Cumulative Frequency Distribution**
- Also known as **ogive**.
22. **Law Equations**
- Probability laws: addition, multiplication, and Bayes' theorem.

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23. **Linear Regression Line**
- A line predicting YY based on XX: Y=a+bXY = a + bX

24. **Y = a + bX, What Does b Represent?**
- bb: **Slope**, showing the change in YY per unit change in XX.

25. **Finding the Median**
- Sort data and find the middle value (odd nn) or average of two middle values (even nn).

26. **Cartesian Product**
- The set of all ordered pairs from two sets. If A={1,2}A = \{1, 2\} and B={a,b}B = \{a, b\}, then A×B={(1,a),(1,b),(2,a),(2,b)}A \times B = \{(1,a), (1,b), (2,a), (2,b)\}.

27. **Draw a Polygon**
- Plot class midpoints (x-axis) against frequencies (y-axis) and connect the points.

28. **First Two Moments of Data**
- **1st Moment:** Mean.
- **2nd Moment:** Variance.

29. **BCR**
- Likely refers to **Benefit-Cost Ratio** in decision-making.

30. **Hypergeometric Distribution**
- Deals with sampling without replacement from a finite population.

31. **Union & Intersection**
- **Union (A∪BA \cup B)**: All elements in AA or BB.
- **Intersection (A∩BA \cap B)**: Common elements in both.

32. **Mean & Standard Deviation for P, N**
- For binomial: μ=np, σ=np(1−p)\mu = np, \, \sigma = \sqrt{np(1-p)}

33. **Binomial Distribution**
- Used when there are fixed trials, two outcomes, and constant success probability.

34. **Playing Cards (Counts)**
- Black: 26; Diamonds: 13; Hearts: 13; Face Cards: 12; Queens: 4.

36. **Continuous Random Topic (Integration)**
- Continuous random variables require integration to calculate probabilities over intervals using their probability density function (PDF).

37. **T-Distribution**
- The T-distribution is used when the sample size is small, and the population variance is unknown. It is symmetric and bell-shaped, but with heavier tails than the normal distribution.

38. **Interval Scale and Ratio Scale (Examples)**
- **Interval Scale:** Temperature in Celsius (no true zero).
- **Ratio Scale:** Weight, height, and age (has a true zero).

39. **Describe Playing Cards**
- A standard deck has 52 cards, divided into 4 suits: spades, hearts, diamonds, and clubs. Each suit has 13 cards, including numbers 2–10, face cards (Jack, Queen, King), and an Ace.

40. **Define Stem-and-Leaf Display**
- A method of organizing data where the stem represents leading digits and the leaf represents trailing digits.

41. **Binomial Experiment Properties**
- Fixed number of trials, two possible outcomes (success/failure), constant probability of success, and independent trials.

42. **ANOVA Table**
- Used in analysis of variance to summarize sources of variation. Key components: between-group variance, within-group variance, F-statistic.

43. **Binomial Distribution**
- A discrete probability distribution for a fixed number of trials with success probability pp. Formula: P(X=k)=(nk)pk(1−p)n−kP(X = k) = \binom{n}{k} p^k (1-p)^{n-k}

44. **How to Find Mean**
- For a dataset: Mean=∑xn\text{Mean} = \frac{\sum x}{n}

45. **Find Cov(X, Y)**
- Formula: Cov(X,Y)=∑(xi−x¯)(yi−y¯)n\text{Cov}(X, Y) = \frac{\sum (x_i - \overline{x})(y_i - \overline{y})}{n}

46. **How to Prove X and Y are Independent**
- Check if P(A∩B)=P(A)P(B)P(A \cap B) = P(A)P(B) or P(X,Y)=P(X)P(Y)P(X, Y) = P(X)P(Y).

47. **Find Population Proportion**
- Formula: p^=x1+x2n1+n2\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} Combine the proportions p1p_1 and p2p_2.

48. **Null and Alternative Hypothesis (Critical Region)**
- Calculate the test statistic. Compare it to the critical value from the statistical table for the significance level α\alpha. If it lies in the critical region, reject H0H₀.

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49. **Confidence Interval (Given S2S^2, nn)**
- Use chi-square distribution for the interval: ((n−1)S2χα/22,(n−1)S2χ1−α/22)\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}} \right)

50. **What About Z-Distribution?**
- A normal distribution with mean 0 and standard deviation 1. Used for hypothesis testing and confidence intervals.

51. **Poisson Distribution**
- Used for counting occurrences over a fixed interval. Formula: P(X=k)=λke−λk!P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}

52. **Mean of Hypergeometric Distribution**
- Formula: μ=nKN\mu = n \frac{K}{N} Where NN: population size, KK: successes in population, nn: sample size.

53. **Normal Distribution**
- A bell-shaped curve where data is symmetrically distributed around the mean. Standard deviation determines the spread.

54. **Binomial Distribution**
- Already explained in Q43.

55. **Chi Distribution**
- A distribution of the square root of a chi-square variable. Used in tests of independence.

56. **Sampling**
- The process of selecting a subset of a population for study.

57. **Testing Hypothesis**
- A method to determine if there is enough evidence to reject the null hypothesis.

58. **Confidence Interval**
- A range of values where the true population parameter is likely to fall.

59. **Degree of Freedom**
- The number of values that are free to vary in calculations.

60. **F-Distribution, T-Test, Z-Test, F-Test**
- **F-Test:** Compares variances.
- **T-Test:** Compares means when variance is unknown.
- **Z-Test:** Compares means when variance is known.

61. **Formula of Z-Test**
- For one sample mean: Z=x¯−μσ/nZ = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}}

62. **Sample Without Replacement Formula**
- Hypergeometric formula: P(X=k)=(Kk)(N−Kn−k)(Nn)P(X=k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}

63. **Definition of Statistics**
- The science of collecting, analyzing, and interpreting data.

64. **Definition of Probability**
- A measure of the likelihood of an event occurring.

65. **Hypothesis by Z-Test**
- Used when testing population means with a known variance.

66. **Estimation**
- A method to infer population parameters from sample data.

67. **Fertile Deviation**
- Possibly refers to **mean deviation**.

68. **Critical Region**
- The area of the distribution where H0H$_0$ is rejected.

69. **Sampling Distribution of Sample Mean**
- Follows normal distribution if sample size is large, mean μ\mu, variance σ2/n\sigma^2/n.

70. **Find Mean & Variance for Given Table**
- Use weighted mean and variance formulas.

71. **Exhaustive Events**
- Events are exhaustive if they cover all possible outcomes of a random experiment. For example, rolling a die has outcomes {1, 2, 3, 4, 5, 6}, which are exhaustive.

72. **Sampling Errors**
- The difference between a sample statistic and the corresponding population parameter due to using only a subset of the population.
    - Causes: Small sample size, biased sampling.
    - Minimized by: Increasing sample size, using random sampling.

73. **Parameters of F-Distribution**
- **Degrees of freedom** for the numerator (df1df_1) and denominator (df2df_2).

74. **How to Find Arithmetic Mean**

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- For ungrouped data: Mean=∑xn\text{Mean} = \frac{\sum x}{n} For grouped data: Mean=∑f·x∑f\text{Mean} = \frac{\sum f \cdot x}{\sum f} Where ff is the frequency, and xx is the class midpoint.

75. **How to Find Mean or Variance**
- **Mean**: Sum of values divided by the number of values.
- **Variance**: Average of squared deviations from the mean. Formula: Variance=∑(x−x¯)2n\text{Variance} = \frac{\sum (x - \overline{x})^2}{n}

76. **Properties of Hypergeometric Distribution**
- No replacement during sampling.
- Probabilities change with each draw.
- Parameters: Population size NN, successes KK, sample size nn.

77. **Construct a Polygon**
- Use midpoints of classes (x-axis) and frequencies (y-axis). Connect points with straight lines.

78. **Find e3x+1e^{3x+1}**
- Compute using the exponential formula.

79. **Find Confidence Limit**
- For a mean: Confidence Interval=x¯±Z·σn\text{Confidence Interval} = \overline{x} \pm Z \cdot \frac{\sigma}{\sqrt{n}} Where ZZ is the critical value, σ\sigma is the standard deviation, nn is the sample size.

80. **Find Coefficient of Variance (C.V.)**
- Formula: C.V.=Standard DeviationMean×100C.V. = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100

81. **Probability of XX Being Even (X=1X = 1 to 14)**
- Possible even numbers: {2, 4, 6, 8, 10, 12, 14}. Total outcomes = 14, Favorable outcomes = 7. P(X is even)=714=0.5P(X \text{ is even}) = \frac{7}{14} = 0.5

82. **Combination**
- Formula for combinations: C(n,r)=n!r!(n−r)!C(n, r) = \frac{n!}{r!(n-r)!} Example: C(5,2)=5!2!·3!=10C(5, 2) = \frac{5!}{2! \cdot 3!} = 10.

83. **Probability Distribution Among 10 Persons**
- Likely uses combinations or binomial distribution to allocate items among persons.

84. **Measures of Dispersion Not Using All Values**
- Examples: Range and quartile deviation.

85. **Two Black Suits in Playing Cards**
- Spades and clubs are black suits.

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